Tu Vu

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Appointments

Virginia Tech — Fall 2024 — *present*
Assistant Professor, Computer Science
Research Interests: *natural language processing & machine learning*

Google
Faculty Researcher — Spring 2025 — *present*
Google DeepMind Research Scientist — 2023 — 2024

Education

University of Massachusetts, Amherst — 2016 — 2023
M.S/Ph.D. in Computer Science
Advisor: Mohit Iyyer — 2018 — 2023
Thesis committee: Mohit Iyyer, Subhransu Maji, Hamed Zamani, Thang Luong, Colin Raffel

Vietnam National University, Hanoi — 2009 — 2013
B.S. Honors Program in Computer Science
Highest distinction (class rank: 1/100)

Professional Experience

Google DeepMind — Fall 2022 — Spring 2023
Student Researcher *with* Thang Luong *&* Quoc Le

Google DeepMind — Summer 2022
Research Intern *with* Thibault Sellam *&* Elizabeth Clark

Google DeepMind — Winter 2021 — Spring 2022
Student Researcher *with* Noah Constant

Google DeepMind — Summer 2021 — Fall 2021
Research Intern *&* Student Researcher *with* Daniel Cer *&* Noah Constant

Google DeepMind — Winter 2020 — Spring 2021
Student Researcher *with* Thang Luong *&* Quoc Le

Google DeepMind — Summer 2020
Research Intern *with* Grady Simon *&* Zi Yang *&* Nan Hua

Microsoft Research — Summer 2019
Research Intern *with* Tong Wang *&* Tsendsuren Munkhdalai *&* Adam Trischler

Selected Preprints & Publications

*: equal contribution
For an up-to-date list of my research papers, please see my Google Scholar profile.

SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models
Thinh Pham, Nguyen Nguyen, Pratibha Zunjare, Weiyuan Chen, Yu-Min Tseng, **Tu Vu**
Under review @ TACL 2025

Efficient Model Development through Fine-tuning Transfer
Pin-Jie Lin, Rishab Balasubramanian, Fengyuan Liu, Nikhil Kandpal, **Tu Vu**
Under review @ EMNLP 2025

What Matters for Model Merging at Scale?
Prateek Yadav, **Tu Vu**, Jonathan Lai, Alexandra Chronopoulou, Manaal Faruqui, Mohit Bansal, Tsendsuren Munkhdalai
Under review @ TMLR 2025

Foundational Autoraters: Taming Large Language Models for Better Automatic Evaluation
Tu Vu*, Kalpesh Krishna*, Salaheddin Alzubi, Chris Tar, Manaal Faruqui, Yun-Hsuan Sung
EMNLP 2024
// The top-performing generative model on RewardBench as of July 15, 2024, trained only on publicly available data

Gemini: A Family of Highly Capable Multimodal Models
Google Gemini Team: Rohan Anil, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew Dai, Anja Hauth, and others including **Tu Vu**
Technical report 2023
// Google AI Blog

FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation
Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong
ACL 2024 Findings
// Our dataset and method have inspired or been used for the development of Google's Gemini, Perplexity.AI's Online LLMs, You.com, and Contextual AI's RAG 2.0

The Flan Collection: Designing Data and Methods for Effective Instruction Tuning
Shayne Longpre, Le Hou, **Tu Vu**, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc Le, Barret Zoph, Jason Wei, and Adam Roberts
ICML 2023
// Google Research Blog

Mixture-of-experts meets instruction tuning: A winning combination for large language models
Sheng Shen, Le Hou, Yanqi Zhou, Nan Du, Shayne Longpre, Jason Wei, Hyung Won Chung, Barret Zoph, William Fedus, Xinyun Chen, **Tu Vu**, Yuexin Wu, Wuyang Chen, Albert Webson, Yunxuan Li, Vincent Zhao, Hongkun Yu, Kurt Keutzer, Trevor Darrell, and Denny Zhou
ICLR 2024

SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer
Tu Vu, Brian Lester, Noah Constant, Rami Al-Rfou, and Daniel Cer
ACL 2022

Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation
Tu Vu, Aditya Barua, Brian Lester, Daniel Cer, Mohit Iyyer, and Noah Constant
EMNLP 2022

STraTA: Self-Training with Task Augmentation for Better Few-shot Learning
Tu Vu, Thang Luong, Quoc Le, Grady Simon, and Mohit Iyyer
EMNLP 2021

Exploring and Predicting Transferability across NLP Tasks
Tu Vu, Tong Wang, Tsendsuren Munkhdalai, Alessandro Sordoni, Adam Trischler, Andrew Mattarella-Micke, Subhransu Maji, and Mohit Iyyer
EMNLP 2020

Funding

VT New Faculty Mentoring Grant (*$1,500*) 2025
PI: Tu Vu

Adobe Research Gift (*$5,000*) 2024
PI: Tu Vu

Advising

PhD advisees:

- Weiyuan Chen, *incoming* PhD student @ **Virginia Tech** — Fall 2025 —
- Jing Chen, *incoming* PhD student @ **Virginia Tech** — Fall 2025 —
- Yu-Min Tseng, *incoming* PhD student @ **Virginia Tech** — Fall 2025 —
- Thinh Pham, 1st year PhD student @ **Virginia Tech** — Fall 2024 — *present*
- Rishab Balasubramanian, 1st year PhD student @ **Virginia Tech** — Fall 2024 — *present*
- Quyet Do, 1st year PhD student @ **Virginia Tech** — Fall 2024 — *present*
- Pin-Jie Lin, 1st year PhD student @ **Virginia Tech** — Fall 2024 — *present*

Masters advisees:

- Noah Provenzano, 1st year MS student @ **Virginia Tech** — Spring 2025 — *present*

Undergrad advisees:

- Rituraj Sharma, senior student @ **Virginia Tech** — Spring 2025 — *present*
- Nguyen Nguyen, sophomore student @ **Virginia Tech** — Spring 2025 — *present*

Others:

- Zhenting Qi, Student Researcher @ **Google** — Summer 2025
- Prateek Yadav, Research Intern @ **Google DeepMind** — Summer 2024 — Spring 2025
- Simeng Han, Student Researcher @ **Google DeepMind** — Summer 2024 — Spring 2025
- Salaheddin Alzubi, Masters student @ **UMass Amherst** — Fall 2022 — Spring 2023
- Dheeraj Mekala, PhD student @ **UCSD** — Spring — Summer 2022

Teaching

CS 4804: Introduction to AI — Fall 2025

CS 5624: Natural Language Processing — Spring 2025

Recent Invited Talks & Lectures

Multimodal LLMs & Test-time Scaling — June 2025
Lecture @ **The New Turing Institute**

Efficient Model Development in the Era of Large Language Models — November 2024
Qualcomm

Efficient Model Development in the Era of Large Language Models — October 2024
Mila / McGill NLP Seminar

Efficient Adaptation of Large Language Models — November 2023
Graph Neural Networks Reading Group, **Google**

Effective and Efficient Transfer Learning in the Era of Large Language Models — Spring 2023
Faculty job talk

Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation — October 2022
Parameter Efficient Tuning Methods Sync, **Google**

Transfer Learning with Large-scale Language Models — August 2022
Lecture @ **The New Turing Institute**

The Appeal of Parameter-efficient Transfer Learning — June 2022
Natural Language Accelerated Team, **Google**

SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer — December 2021
Parameter Efficient Tuning Methods Sync, **Google**

Academic Service

Area Chair for ACL 2025; NAACL 2025; COLING 2025; ACL 2024; EMNLP 2025, 2024

Session Chair (Machine Learning for NLP) @ EMNLP 2024

Program Committee/Reviewer for Nature 2025, ICML 2025; NeurIPS 2024; TL4NLP@NeurIPS 2022; MCDC@ICLR 2025; COLM 2024; AAAI 2023; ACL 2022, 2021, 2020, 2019; EMNLP 2022, 2021; NAACL 2022, 2021; COLING 2020; CoNLL 2019; INLG 2020, 2019

Selected Media

Gemini: Google AI Blog — 2023

FreshLLMs: ZDNET — 2023

The Flan Collection: Google Research Blog — 2023

SPoT: Headlines of Google AI's Natural Language Accelerated Newsletter — Q1, 2022

Selected Awards & Honors

Google Student Researcherships	2020 — 2023
UMass Amherst Graduate Assistantships	2016 — 2023
Honda Y-E-S Award for young engineers and scientists, Vietnam *// in the top 10 nationally*	2013
Outstanding Academic and Co-curricular Achievements, Vietnam National University	2013
Prominent Young Figure Award, Vietnam National University	2010 & 2012
First Runner-up Prize, International Programming Contest, Japan *// ranked 2nd among 64 teams internationally*	2011
Outstanding Young Talent of the Capital City, Vietnam *// in the top 100 most outstanding young talents selected from a wide range of fields*	2010
Champion Prize, National Mathematical Olympiad, Vietnam *// ranked 1st among more than 600 contestants nationally*	2010
A number of prizes in National/International Olympiads (in both Mathematics and Informatics)	2009 — 2013
A number of academic scholarships for undergraduate students	2009 — 2013

Patents

*: original inventor

Frozen Model Adaptation Through Soft Prompt Transfer
Tu Vu*, Daniel Cer, Noah Constant, Brian Lester, Rami Al-Rfou
U.S. Patent Application, 17/863,840

Task Augmentation and Self-training for Improved Few-shot Learning
Thang Luong, **Tu Vu***, Quoc Le, Grady Simon
U.S. Patent Application, 17/826,690